Compton Scattering Cross Section in QED: From Scratch, Every Step

Griffiths 7.30 + 7.41 | Professor's Convention | H&M Gauge | Mandelstam

For Highly Relativistic Particles: Drop All Terms $\propto m$

PHYS 7502 — Particles and Nuclei 2

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Part I
Background: Tools We Need

1 Why We Drop Mass Terms

Highly Relativistic Limit

The problem says: **do NOT show terms proportional to mass**. This means we set $m = 0$ for the electron everywhere.
Why this makes sense: At high energies $E \gg mc^2$, the electron mass is negligible compared to its momentum. Every mass insertion in a propagator or trace produces terms suppressed by powers of m/E. Dropping them simplifies the algebra enormously while capturing the dominant physics.
What changes at $m = 0$:

- On-shell: $p^2 = m^2 \to 0$ for all external fermions.
- Propagator numerator: $\not{q} + m \to \not{q}$.
- Completeness: $\sum u\bar{u} = \not{p} + m \to \not{p}$.
- Mandelstam constraint: $s + t + u = 2m^2 \to 0$.
- Dirac equation on external spinors: $\not{p}\, u(p) = mu(p) \to 0$.

2 QED Feynman Rules (What We Use)

Rules for Building Amplitudes (Griffiths §7.6)

1. **External electron:** incoming $\to u(p)$; outgoing $\to \bar{u}(p)$.
2. **External photon:** incoming $\to \varepsilon_\mu(p)$; outgoing $\to \varepsilon^*_\mu(p)$.
3. **QED vertex:** $ig_e\gamma^\mu$ (index μ contracts with the photon at that vertex).
4. **Internal fermion propagator** (momentum q): $\dfrac{i\not{q}}{q^2}$ (at $m = 0$).
5. **Momentum conservation:** $(2\pi)^4\delta^{(4)}(\sum p_{\text{in}} - \sum p_{\text{out}})$ at each vertex.
6. **Internal momentum integral:** $\int d^4q/(2\pi)^4$ per internal line.
7. **Reading order:** Follow each fermion line *backward* (against the arrow), writing $\bar{u} \cdots [\text{vertex}] \cdots [\text{propagator}] \cdots [\text{vertex}] \cdots u$.

3 Casimir's Trick (Griffiths Eq. 7.125)

Casimir's Trick: Spin Sums → Traces

For any 4×4 matrices Γ_1, Γ_2, and at $m = 0$:

$$\sum_{\text{all spins}} \Big[\bar{u}(a)\,\Gamma_1\,u(b)\Big]\Big[\bar{u}(b)\,\bar{\Gamma}_2\,u(a)\Big] = \mathrm{Tr}\Big[\Gamma_1\,\not{p}_b\,\Gamma_2\,\not{p}_a\Big] \tag{1}$$

where $\bar{\Gamma} \equiv \gamma^0\Gamma^\dagger\gamma^0$ reverses the order of the γ-matrices in Γ.
How it works:

1. Completeness $\sum_s u^{(s)}(b)\bar{u}^{(s)}(b) = \not{p}_b$ stitches $u(b)$ and $\bar{u}(b)$ into $\not{p}_b$.
2. Completeness $\sum_s u^{(s)}(a)\bar{u}^{(s)}(a) = \not{p}_a$ closes the loop.
3. What remains is a trace of 4×4 matrices — no spinors left.

4 Photon Polarization Sum (H&M Eq. 6.111)

$-g_{\mu\nu}$ Replacement

The Ward identity ($k^\mu\mathcal{M}_\mu = 0$) guarantees that unphysical polarizations decouple from gauge-invariant amplitudes. Therefore:

$$\sum_{\lambda=1,2} \varepsilon_\mu^{(\lambda)}(k)\,\varepsilon_\nu^{(\lambda)*}(k) \;\longrightarrow\; -g_{\mu\nu} \tag{2}$$

This replaces Griffiths' Coulomb gauge form $\delta_{ij} - \hat{k}_i\hat{k}_j$.

5 Contraction and Trace Theorems

The Identities We Will Use (Griffiths §7.7)

Contraction theorems (4 dimensions):

$$\text{Thm 6:}\quad \gamma^\mu\gamma_\mu = 4 \tag{3}$$
$$\text{Thm 7:}\quad \gamma^\mu\not{a}\,\gamma_\mu = -2\not{a} \tag{4}$$
$$\text{Thm 8}'\text{:}\quad \gamma^\mu\not{a}\,\not{b}\,\gamma_\mu = 4(a\cdot b) \tag{5}$$
$$\text{Thm 9:}\quad \gamma^\mu\not{a}\,\not{b}\,\not{c}\,\gamma_\mu = -2\not{c}\,\not{b}\,\not{a} \tag{6}$$

Trace theorems:

$$\text{Thm 10:}\quad \mathrm{Tr}(\text{odd \# of }\gamma\text{'s}) = 0 \tag{7}$$
$$\text{Thm 11:}\quad \mathrm{Tr}(\mathbf{1}) = 4 \tag{8}$$
$$\text{Thm 12}'\text{:}\quad \mathrm{Tr}(\not{a}\,\not{b}) = 4\,a\cdot b \tag{9}$$
$$\text{Thm 13}'\text{:}\quad \mathrm{Tr}(\not{a}\,\not{b}\,\not{c}\,\not{d}) = 4\big[(a\cdot b)(c\cdot d) - (a\cdot c)(b\cdot d) + (a\cdot d)(b\cdot c)\big] \tag{10}$$

Anticommutation:

$$\not{a}\not{b} + \not{b}\not{a} = 2\,a \cdot b \qquad \Rightarrow \qquad \not{a}\not{b} = 2\,a \cdot b - \not{b}\not{a} \tag{11}$$

Massless on-shell:

$$\not{p}^2 = p^2 = 0 \quad \text{for all external particles at } m = 0 \tag{12}$$

6 Mandelstam Variables

Mandelstam Variables (Professor's Convention)

For $e^-(p_1) + \gamma(p_2) \to \gamma(p_3) + e^-(p_4)$:

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2 \tag{13}$$
$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2 \tag{14}$$
$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2 \tag{15}$$

At $m = 0$: $s + t + u = 0$, and all dot products become:

$$\boxed{p_1 \cdot p_2 = \frac{s}{2}, \quad p_1 \cdot p_3 = -\frac{t}{2}, \quad p_1 \cdot p_4 = -\frac{u}{2}, \quad p_3 \cdot p_4 = \frac{s}{2}, \quad p_2 \cdot p_4 = -\frac{t}{2}, \quad p_2 \cdot p_3 = -\frac{u}{2}} \tag{16}$$

Part II
The Calculation

7 Step 1: Identify the Diagrams

For $e^-(p_1) + \gamma(p_2) \to \gamma(p_3) + e^-(p_4)$, there are exactly **two** tree-level diagrams. They **add** (no relative minus sign—no fermion interchange).

Diagram s: s-channel	**Diagram t: t-channel**
e^- absorbs $\gamma_{\text{in}}(p_2)$ first	e^- emits $\gamma_{\text{out}}(p_3)$ first
Virtual e^- momentum: $q = p_1 + p_2$	Virtual e^- momentum: $q = p_1 - p_3$
Then emits $\gamma_{\text{out}}(p_3)$	Then absorbs $\gamma_{\text{in}}(p_2)$
Propagator denominator: $(p_1 + p_2)^2 = s$	Propagator denominator: $(p_1 - p_3)^2 = t$

8 Step 2: Write the Full Integrals

8.1 s-channel integral

Reading backward along the fermion line ($\bar{u}(4) \leftarrow \cdots \leftarrow u(1)$):

$$\begin{aligned} & i\mathcal{M}_s \times (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) \\ &= \int \frac{d^4q}{(2\pi)^4} \left[\bar{u}(4)\,(ig_e\gamma^\alpha)\,\frac{i\not{q}}{q^2}\,(ig_e\gamma^\beta)\,u(1)\right] \varepsilon^*_{3\alpha}\,\varepsilon_{2\beta} \\ &\quad \times (2\pi)^4 \delta^{(4)}(q - p_3 - p_4) \times (2\pi)^4 \delta^{(4)}(p_1 + p_2 - q) \end{aligned} \tag{17}$$

Reading guide for the spinor sandwich:

$\bar{u}(4)$	outgoing electron exits
$(ig_e\gamma^\alpha)$	Vertex B: emits $\gamma_{\text{out}}(p_3)$; index α contracts with $\varepsilon^*_{3\alpha}$
$i\not{q}/(q^2)$	virtual electron propagator
$(ig_e\gamma^\beta)$	Vertex A: absorbs $\gamma_{\text{in}}(p_2)$; index β contracts with $\varepsilon_{2\beta}$
$u(1)$	incoming electron enters

8.2 Performing the q integration

The δ-function $\delta^{(4)}(p_1+p_2-q)$ sets $q = p_1+p_2$. The remaining δ-function becomes overall conservation. Stripping it off and simplifying $(ig_e)^2 \cdot i = -ig_e^2$:

$\mathcal{M}_s$ (Professor's Convention, $m = 0$)

$$\boxed{\mathcal{M}_s = \frac{e^2}{s}\,\varepsilon^*_{3\alpha}\,\varepsilon_{2\beta}\,\bar{u}(4)\,\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\,u(1)} \tag{18}$$

where we used $g_e^2 = e^2$ and $(p_1 + p_2)^2 = s$.

8.3 t-channel integral

Same procedure. The δ-function at the first vertex sets $q = p_1 - p_3$:

$\mathcal{M}_t$ (Professor's Convention, $m = 0$)

$$\boxed{\mathcal{M}_t = \frac{e^2}{t}\,\varepsilon_{2\mu}\,\varepsilon^*_{3\nu}\,\bar{u}(4)\,\gamma^\mu(\not{p}_1 - \not{p}_3)\gamma^\nu\,u(1)} \tag{19}$$

Note the polarizations appear in **swapped order** vs. $\mathcal{M}_s$: here ε_2 (incoming) appears with the vertex where γ_{in} is absorbed, and ε_3^* (outgoing) with the emission vertex. This reflects the reversed time-ordering of the photon interactions.

Total: $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t$.

9 Step 3: Write $|\mathcal{M}|^2$ (Four Terms)

$$|\mathcal{M}|^2 = |\mathcal{M}_s|^2 + |\mathcal{M}_t|^2 + \mathcal{M}_s\mathcal{M}_t^* + \mathcal{M}_s^*\mathcal{M}_t \tag{20}$$

We compute each term separately, then average:

$$\overline{|\mathcal{M}|^2} = \frac{1}{4}\sum_{\text{electron spins}}\sum_{\text{photon pol.}}|\mathcal{M}|^2$$

The $\frac{1}{4}$ averages over 2 initial electron spins $\times$ 2 initial photon polarizations.

10 Step 4: Conjugating the Amplitudes

To form $|\mathcal{M}_s|^2 = \mathcal{M}_s\mathcal{M}_s^*$ and $\mathcal{M}_s\mathcal{M}_t^*$, we need $\mathcal{M}_s^*$ and $\mathcal{M}_t^*$.

Using $[\bar{u}(a)\Gamma\,u(b)]^* = \bar{u}(b)\bar{\Gamma}\,u(a)$ where $\bar{\Gamma}$ reverses the γ-matrix order and conjugates polarization vectors:

$$\mathcal{M}_s^* = \frac{e^2}{s}\,\varepsilon_{3\alpha}\,\varepsilon^*_{2\beta}\,\bar{u}(1)\,\gamma^\beta(\not{p}_1 + \not{p}_2)\gamma^\alpha\,u(4) \tag{21}$$

$$\mathcal{M}_t^* = \frac{e^2}{t}\,\varepsilon^*_{2\mu}\,\varepsilon_{3\nu}\,\bar{u}(1)\,\gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu\,u(4) \tag{22}$$

11 Step 5: Casimir's Trick for $|\mathcal{M}_s|^2$

11.1 Form the product

$$|\mathcal{M}_s|^2 = \frac{e^4}{s^2}\,\varepsilon^*_{3\alpha}\varepsilon_{3\rho}\,\varepsilon_{2\beta}\varepsilon^*_{2\sigma} \times \underbrace{\bar{u}(4)\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta u(1)}_{A}\,\underbrace{\bar{u}(1)\gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho u(4)}_{B} \tag{23}$$

11.2 Identify the Casimir structure

$A = \bar{u}(4)\Gamma_1 u(1)$, $B = \bar{u}(1)\Gamma_2 u(4)$. This has the a-b-b-a pattern with $a = 4$, $b = 1$:

$$\Gamma_1 = \gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta \tag{24}$$

$$\Gamma_2 = \gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho \tag{25}$$

11.3 Apply Casimir's trick

$$\sum_{\text{spins}} A \cdot B = \mathrm{Tr}\Big[\underbrace{\gamma^\alpha(\not{p}_1+\not{p}_2)\gamma^\beta}_{\Gamma_1}\ \underbrace{\not{p}_1}_{\text{particle 1}}\ \underbrace{\gamma^\sigma(\not{p}_1+\not{p}_2)\gamma^\rho}_{\Gamma_2}\ \underbrace{\not{p}_4}_{\text{particle 4}}\ \Big] \tag{26}$$

11.4 Apply photon polarization sums

$\sum \varepsilon^*_{3\alpha}\varepsilon_{3\rho} \to -g_{\alpha\rho}$ contracts α with ρ.
$\sum \varepsilon_{2\beta}\varepsilon^*_{2\sigma} \to -g_{\beta\sigma}$ contracts β with σ.
Two factors of (-1) give $+1$:

After Spin + Polarization Sums (s-channel)

$$\frac{1}{4}\sum |\mathcal{M}_s|^2 = \frac{e^4}{4s^2}\ \mathrm{Tr}\Big[\gamma^\alpha(\not{p}_1+\not{p}_2)\gamma^\beta\ \not{p}_1\ \gamma_\beta(\not{p}_1+\not{p}_2)\gamma_\alpha\ \not{p}_4\Big] \tag{27}$$

This is a trace of **8 gamma matrices** (each $\not{p}$ counts as 1).

12 Step 6: Evaluate the s-Channel Trace

12.1 Inner contraction: $\gamma^\beta\,\not{p}_1\,\gamma_\beta$

By Theorem 7 (4): $\gamma^\beta \not{a} \gamma_\beta = -2\not{a}$.

$$\gamma^\beta\,\not{p}_1\,\gamma_\beta = -2\not{p}_1 \tag{28}$$

Substituting into (27):

$$= \frac{e^4}{4s^2}\,(-2)\ \mathrm{Tr}\Big[\gamma^\alpha(\not{p}_1+\not{p}_2)\ \not{p}_1\ (\not{p}_1+\not{p}_2)\gamma_\alpha\ \not{p}_4\Big] \tag{29}$$

12.2 Central product: $(\not{p}_1+\not{p}_2)\,\not{p}_1\,(\not{p}_1+\not{p}_2)$

Expand, using $\not{p}_1^2 = p_1^2 = 0$:

$$\begin{aligned}(\not{p}_1+\not{p}_2)\not{p}_1(\not{p}_1+\not{p}_2) &= \underbrace{\not{p}_1\not{p}_1}_{=0}(\not{p}_1+\not{p}_2) + \not{p}_2\not{p}_1(\not{p}_1+\not{p}_2)\\ &= \not{p}_2\underbrace{\not{p}_1\not{p}_1}_{=0} + \not{p}_2\not{p}_1\not{p}_2\\ &= \not{p}_2\not{p}_1\not{p}_2\end{aligned} \tag{30}$$

Now simplify $\not{p}_2\not{p}_1\not{p}_2$ using anticommutation (11):

$$\begin{aligned}&\not{p}_2\not{p}_1 = 2(p_2\cdot p_1) - \not{p}_1\not{p}_2\\ \Rightarrow\quad &\not{p}_2\not{p}_1\not{p}_2 = 2(p_1\cdot p_2)\not{p}_2 - \not{p}_1\underbrace{\not{p}_2^2}_{=0} = 2(p_1\cdot p_2)\,\not{p}_2\end{aligned} \tag{31}$$

Using $p_1 \cdot p_2 = s/2$:

Key Identity (s-channel)

$$\boxed{(\not{p}_1 + \not{p}_2)\,\not{p}_1\,(\not{p}_1 + \not{p}_2) = s\,\not{p}_2} \tag{32}$$

Substituting into (29):

$$= \frac{e^4}{4s^2}\,(-2)(s)\,\mathrm{Tr}\Big[\gamma^\alpha\,\not{p}_2\,\gamma_\alpha\,\not{p}_4\Big] \tag{33}$$

12.3 Outer contraction: $\gamma^\alpha\,\not{p}_2\,\gamma_\alpha$

By Theorem 7: $\gamma^\alpha \not{p}_2 \gamma_\alpha = -2\not{p}_2$.

$$= \frac{e^4}{4s^2}\,(-2)(s)(-2)\,\mathrm{Tr}\Big[\not{p}_2\,\not{p}_4\Big] = \frac{e^4}{s}\,\mathrm{Tr}(\not{p}_2\not{p}_4) \tag{34}$$

12.4 Final trace: $\mathrm{Tr}(\not{p}_2\not{p}_4)$

By Theorem 12′ (9): $\mathrm{Tr}(\not{a}\not{b}) = 4\,a\cdot b$.

$$\mathrm{Tr}(\not{p}_2\not{p}_4) = 4\,p_2\cdot p_4 = 4\left(-\frac{t}{2}\right) = -2t \tag{35}$$

s-Channel Result

$$\boxed{\frac{1}{4}\sum_{\text{spins,pol}} |\mathcal{M}_s|^2 = \frac{e^4}{s}(-2t) = -\frac{2e^4\,t}{s}} \tag{36}$$

13 Step 7: Casimir's Trick for $|\mathcal{M}_t|^2$

13.1 Form the product, apply Casimir + photon sums

By identical reasoning ($\Gamma_1 = \gamma^\mu(\not{p}_1 - \not{p}_3)\gamma^\nu$, $\Gamma_2 = \gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu$ after contraction):

$$\frac{1}{4}\sum |\mathcal{M}_t|^2 = \frac{e^4}{4t^2}\,\mathrm{Tr}\Big[\gamma^\mu(\not{p}_1 - \not{p}_3)\gamma^\nu\,\not{p}_1\,\gamma_\nu(\not{p}_1 - \not{p}_3)\gamma_\mu\,\not{p}_4\Big] \tag{37}$$

14 Step 8: Evaluate the t-Channel Trace

14.1 Inner contraction

$\gamma^\nu \not{p}_1 \gamma_\nu = -2\not{p}_1$ (same Theorem 7).

14.2 Central product: $(\not{p}_1 - \not{p}_3)\,\not{p}_1\,(\not{p}_1 - \not{p}_3)$

$$\begin{aligned}(\not{p}_1 - \not{p}_3)\not{p}_1(\not{p}_1 - \not{p}_3) &= -\not{p}_3\not{p}_1(\not{p}_1 - \not{p}_3) \qquad (\not{p}_1^2 = 0)\\ &= -\not{p}_3\underbrace{\not{p}_1^2}_{=0} + \not{p}_3\not{p}_1\not{p}_3 = \not{p}_3\not{p}_1\not{p}_3\end{aligned} \tag{38}$$

Anticommute: $\not{p}_3\not{p}_1 = 2(p_3\cdot p_1) - \not{p}_1\not{p}_3$, then $\not{p}_3^2 = 0$:

$$\not{p}_3\not{p}_1\not{p}_3 = 2(p_1\cdot p_3)\not{p}_3 = 2\left(-\frac{t}{2}\right)\not{p}_3 = (-t)\,\not{p}_3 \tag{39}$$

Key Identity (t-channel)

$$\boxed{(\not{p}_1 - \not{p}_3)\,\not{p}_1\,(\not{p}_1 - \not{p}_3) = (-t)\,\not{p}_3} \tag{40}$$

14.3 Outer contraction + trace

$$\begin{aligned}\frac{e^4}{4t^2}(-2)(-t)\ \mathrm{Tr}\Big[\gamma^\mu \not{p}_3 \gamma_\mu\, \not{p}_4\Big] &= \frac{e^4}{4t^2}(-2)(-t)(-2)\ \mathrm{Tr}(\not{p}_3\not{p}_4)\\ &= \frac{e^4}{4t^2}\,(4t)\ \cdot\ 4(p_3\cdot p_4)\end{aligned} \tag{41}$$

Using $p_3\cdot p_4 = s/2$:

$$= \frac{e^4}{4t^2}\,(4t)\ \cdot\ 4\left(\frac{s}{2}\right) = \frac{e^4}{4t^2}\cdot 4t\cdot 2s = \frac{e^4}{t}\cdot 2s\cdot\frac{(-1)\cdot(-1)}{1} \tag{42}$$

Let me redo this carefully:

$$\begin{aligned}&\frac{e^4}{4t^2}\ \times\ (-2)\ \times\ (-t)\ \times\ (-2)\ \times\ \mathrm{Tr}(\not{p}_3\not{p}_4)\\ &= \frac{e^4}{4t^2}\ \times\ (-4t)\ \times\ (4\cdot p_3\cdot p_4)\\ &= \frac{e^4}{4t^2}\ \times\ (-4t)\ \times\ \left(4\cdot\frac{s}{2}\right)\\ &= \frac{e^4}{4t^2}\ \times\ (-4t)\ \times\ (2s)\\ &= \frac{e^4}{4t^2}\ \times\ (-8st)\\ &= -\frac{2e^4 s}{t}\end{aligned} \tag{43}$$

t-Channel Result

$$\boxed{\frac{1}{4}\sum_{\text{spins,pol}}|\mathcal{M}_t|^2 = -\frac{2e^4\,s}{t}} \tag{44}$$

15 Step 9: The Interference Term $\mathcal{M}_s\mathcal{M}_t^*$

15.1 Form the product and apply Casimir + photon sums

Following the same Casimir procedure (now with Γ_1 from the s-channel, Γ_2 from the t-channel):

$$\frac{1}{4}\sum \mathcal{M}_s\mathcal{M}_t^* = \frac{e^4}{4st}\ \mathrm{Tr}\Big[\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\ \not{p}_1\ \gamma_\alpha(\not{p}_1 - \not{p}_3)\gamma_\beta\ \not{p}_4\Big] \tag{45}$$

15.2 Contract α (Theorem 9)

$\gamma^\alpha(\not{p}_1+\not{p}_2)\gamma^\beta\not{p}_1\gamma_\alpha$: this is $\gamma^\alpha\not{a}\not{b}\not{c}\gamma_\alpha$ with $\not{a}=\not{p}_1+\not{p}_2$, $\not{b}=\gamma^\beta$, $\not{c}=\not{p}_1$.

By Theorem 9 (6): $\gamma^\alpha\not{a}\not{b}\not{c}\gamma_\alpha=-2\not{c}\not{b}\not{a}$.

$$\gamma^\alpha(\not{p}_1+\not{p}_2)\gamma^\beta\not{p}_1\gamma_\alpha=-2\,\not{p}_1\gamma^\beta(\not{p}_1+\not{p}_2) \tag{46}$$

Substitute:

$$=\frac{e^4}{4st}\,(-2)\,\mathrm{Tr}\Big[\not{p}_1\gamma^\beta\underbrace{(\not{p}_1+\not{p}_2)(\not{p}_1-\not{p}_3)}_{\text{expand}}\gamma_\beta\,\not{p}_4\Big] \tag{47}$$

15.3 Expand $(\not{p}_1+\not{p}_2)(\not{p}_1-\not{p}_3)$

$$(\not{p}_1+\not{p}_2)(\not{p}_1-\not{p}_3)=\underbrace{\not{p}_1^2}_{=0}-\not{p}_1\not{p}_3+\not{p}_2\not{p}_1-\not{p}_2\not{p}_3 \tag{48}$$

15.4 Contract β on each term (Theorem 8′)

By Theorem 8′ (5): $\gamma^\beta\not{a}\not{b}\gamma_\beta=4(a\cdot b)$.

$$\gamma^\beta(-\not{p}_1\not{p}_3)\gamma_\beta=-4\,(p_1\cdot p_3)=-4\Big(-\frac{t}{2}\Big)=+2t \tag{49}$$

$$\gamma^\beta(\not{p}_2\not{p}_1)\gamma_\beta=+4\,(p_2\cdot p_1)=+4\Big(\frac{s}{2}\Big)=+2s \tag{50}$$

$$\gamma^\beta(-\not{p}_2\not{p}_3)\gamma_\beta=-4\,(p_2\cdot p_3)=-4\Big(-\frac{u}{2}\Big)=+2u \tag{51}$$

Sum:

$$2t+2s+2u=2(s+t+u)=2\times 0=0 \tag{52}$$

Interference Vanishes

$$\boxed{\frac{1}{4}\sum\mathcal{M}_s\mathcal{M}_t^*=0\qquad(\text{and likewise }\mathcal{M}_s^*\mathcal{M}_t=0)} \tag{53}$$

The zero comes from $s+t+u=0$ at $m=0$. At $m\neq 0$ this would give $2(2m^2)=4m^2$, a nonzero but small correction.

16 Step 10: Combine Everything

From (36), (44), (53):

FINAL RESULT: Compton $\overline{|\mathcal{M}|^2}$ in Mandelstam Variables

$$\boxed{\overline{|\mathcal{M}|^2}=-\frac{2e^4\,t}{s}-\frac{2e^4\,s}{t}+0+0=-2e^4\Big(\frac{t}{s}+\frac{s}{t}\Big)=-2e^4\,\frac{s^2+t^2}{st}} \tag{54}$$

Positivity: $s>0$, $t<0\Rightarrow -t/s>0$, $-s/t>0\Rightarrow\overline{|\mathcal{M}|^2}>0$. ✓

This matches H&M Eq. post-6.112 and L&B Eq. (40.16) (with the convention translation $t_{\text{here}}=u_{\text{L\&B}}$).

Part III
Extensions

17 Invariant Cross Section $d\sigma/dt$

For massless $2 \to 2$: $d\sigma/dt = \overline{|\mathcal{M}|^2}/(16\pi s^2)$. With $e^4 = 16\pi^2\alpha^2$:

Compton $d\sigma/dt$

$$\boxed{\frac{d\sigma}{dt} = -\frac{2\pi\alpha^2(s^2+t^2)}{s^3\,t}} \tag{55}$$

18 Crossing to Pair Annihilation (Problem 7.41)

Crossing: $p_2 \to -q_3$, $p_4 \to -q_2$ maps $s_C \to t_A$, $t_C \to u_A$, $u_C \to s_A$.

Pair Annihilation $e^+e^- \to \gamma\gamma$

$$\boxed{\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma} = -2e^4\left(\frac{u}{t}+\frac{t}{u}\right)}, \qquad \frac{d\sigma}{dt} = -\frac{2\pi\alpha^2(t^2+u^2)}{s^2\,tu} \tag{56}$$

19 Virtual Photon (H&M Exercise 6.19)

If $p_2^2 = -Q^2 \neq 0$: the key identity (32) changes because $\not{p}_2^2 = -Q^2$:

$$\not{p}_2\not{p}_1\not{p}_2 = 2(p_1\cdot p_2)\not{p}_2 - \not{p}_1\not{p}_2^2 = (s+Q^2)\not{p}_2 + Q^2\not{p}_1$$

The extra term generates corrections, and the interference no longer vanishes ($s+t+u = -Q^2 \neq 0$):

Virtual Photon Compton

$$\boxed{\overline{|\mathcal{M}|^2} = 2e^4\left(-\frac{t}{s}-\frac{s}{t}+\frac{2Q^2u}{st}\right)} \tag{57}$$

Appendix: Quick Reference

All Identities Used

$\gamma^\mu \not{a}\gamma_\mu = -2\not{a}$. $\quad \gamma^\mu \not{a}\not{b}\gamma_\mu = 4(a\cdot b)$. $\quad \gamma^\mu \not{a}\not{b}\not{c}\gamma_\mu = -2\not{c}\not{b}\not{a}$.
$\mathrm{Tr}(\not{a}\not{b}) = 4\,a\cdot b$. $\quad \not{a}\not{b} = 2a\cdot b - \not{b}\not{a}$. $\quad \not{p}^2 = p^2 = 0$ (massless).
$\sum u\bar{u} = \not{p}$. $\quad \sum \varepsilon_\mu\varepsilon_\nu^* \to -g_{\mu\nu}$. $\quad s+t+u = 0$.
$p_1\cdot p_2 = s/2$. $\quad p_1\cdot p_3 = -t/2$. $\quad p_1\cdot p_4 = -u/2$. $\quad p_2\cdot p_4 = -t/2$. $\quad p_3\cdot p_4 = s/2$. $\quad p_2\cdot p_3 = -u/2$.